

**LABORATORY MANUAL CUM OBSERVATION BOOK**  
***BASIC ELECTRICAL ENGINEERING LAB***  
**(CODE: ES354EE)**

For

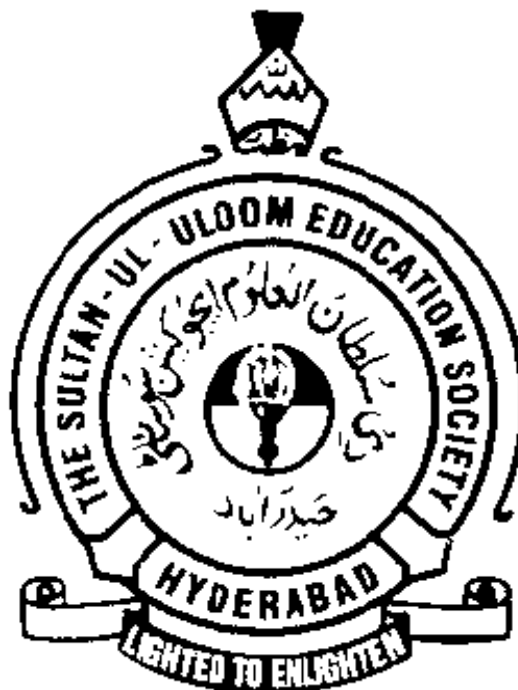
**B.E. I/IV (AICTE Model Curriculum)**

Branch: \_\_\_\_\_

Semester: \_\_\_\_\_

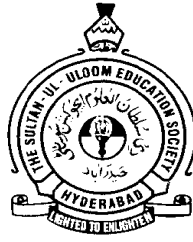
Academic Year: \_\_\_\_\_

**DEPARTMENT OF ELECTRICAL ENGINEERING**



**MUFFAKHAM JAH COLLEGE OF ENGINEERING & TECHNOLOGY**  
**Affiliated to Osmania University, Accredited by NAAC-A+ & NBA**

Mount Pleasant, 8-2-249 to 267, Road No. 3, Banjara Hills, Hyderabad - 500  
034, Telangana State, India +91 4023280301, [www.mjcollege.ac.in](http://www.mjcollege.ac.in)



## MUFFAKHAM JAH COLLEGE OF ENGINEERING & TECHNOLOGY

Affiliated to Osmania University, Accredited by NAAC-A+ & NBA  
Mount Pleasant, 8-2-249 to 267, Road No. 3, Banjara Hills, Hyderabad - 500 034, Telangana State,  
India +91 4023280301, [www.mjcollege.ac.in](http://www.mjcollege.ac.in)

### DEPARTMENT OF ELECTRICAL ENGINEERING

#### CERTIFICATE

This is to certify that this observation book is a bonafide record of practical work done in **Basic Electrical Engineering Lab** in BE.\_\_\_\_\_, I-Year, \_\_\_\_-Sem, during the Academic Year\_\_\_\_\_

**Name** : \_\_\_\_\_  
**Roll No** : \_\_\_\_\_  
**Batch** : \_\_\_\_\_

COURSE COORDINATOR

**INDEX**  
**Cycle-I**

| Expt No. | Date | Name of the Experiment                                                                                                                                                                                                                              | Marks (20M) | Signature |
|----------|------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|-----------|
| 1.       |      | a) Verification of KVL<br>b) Verification of KCL<br>c) Verification of super position theorem                                                                                                                                                       |             |           |
| 2.       |      | a) Verification of Thevenin's theorem<br>b) Verification of Norton's theorem                                                                                                                                                                        |             |           |
| 3.       |      | a) Sinusoidal steady state response of RL series circuits<br>b) Sinusoidal steady state response of RC series circuits                                                                                                                              |             |           |
| 4.       |      | a) Measurement of phase voltage/current, line voltage/current and power in balanced three-phase star connected circuit.<br>b) Measurement of phase voltage/current, line voltage/current and power in balanced three-phase delta connected circuit. |             |           |

**Cycle-II**

| Expt No | Date | Name of the Experiment                                                                                                                                                            | Marks (20M) | Signature |
|---------|------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|-----------|
| 5.      |      |                                                                                                                                                                                   |             |           |
|         |      | Loading of single phase transformer                                                                                                                                               |             |           |
| 6.      |      | (a). Synchronous speed & speed of rotor for two and four- pole,3- $\phi$ induction motor.<br>(b) Direction reversal by change of phase-sequence of 3-phase ac supply connections. |             |           |
| 7.      |      | Power factor improvement of Single Phase induction motor using static capacitors                                                                                                  |             |           |
| 8.      |      | .Magnetization curve of a separately excited DC generator                                                                                                                         |             |           |

**Note:**

\*Print out should be taken on only one side

\*Theoretical Calculations should be on Left hand side

\*Every week Experimentation Observations and Calculations are assessed for 20M



**MUFFAKHAM JAH COLLEGE OF ENGINEERING TECHNOLOGY**  
**ELECTRICAL ENGINEERING DEPARTMENT**  
**CONDUCT OF EXPERIMENT SCHEDULE CYCLE-I**

Subject: **Basic Electrical Engineering Lab**

Class: BE 1/4, \_\_\_-\_\_\_ (\_\_\_ – Sem)

Subject Code: ES354EE

Faculty Name:

**Scheduled Lab Batches : A/B/C** (1604-\_\_-\_\_-\_\_ TO 1604-\_\_-\_\_-\_\_)

| Sl. No | Date on conduction                                                                                                                                                 | WK-1 | WK-2 | WK-3 | WK-4 |
|--------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|------|------|------|
|        | Name of Experiment                                                                                                                                                 |      |      |      |      |
| 1      | a) Verification of KVL<br>b) Verification of KCL<br>c) Verification of super position theorem                                                                                                                                                       | A    | D    | C    | B    |
| 2      | a) Verification of Thevenin's theorem<br>b) Verification of Norton's theorem                                                                                                                                                                        | B    | A    | D    | C    |
| 3      | a) Sinusoidal steady state response of RL series circuits<br>b) Sinusoidal steady state response of RC series circuits                                                                                                                              | C    | B    | A    | D    |
| 4      | a) Measurement of phase voltage/current, line voltage/current and power in balanced three-phase star connected circuit.<br>b) Measurement of phase voltage/current, line voltage/current and power in balanced three-phase delta connected circuit. | D    | C    | B    | A    |

**Lab Internal Exam:** ( 1 ONLY)**Student Groups:****Course Coordinator:**

1604-\_\_-\_\_-\_\_

A:

B:

C:

D:



**MUFFAKHAM JAH COLLEGE OF ENGINEERING TECHNOLOGY**  
**ELECTRICAL ENGINEERING DEPARTMENT**  
**CONDUCT OF EXPERIMENT SCHEDULE-CYCLE-II**

Subject: **Basic Electrical Engineering Lab**

Class: BE 1/4, \_\_-\_\_ ( \_\_ – Sem)

Subject Code: ES354EE

Faculty Name:

**Scheduled Lab Batches : A/B/C** (1604-\_\_-\_\_-\_\_ T0 1604-\_\_-\_\_-\_\_)

| Sl. No | Date on conduction                                                                                | WK-1 | WK-2 | WK-3 | WK-4 |
|--------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|------|------|------|
|        | Name of Experiment                                                                                |      |      |      |      |
| 5      | Loading of single phase transformer                                                                                                                                                | A    | D    | C    | B    |
| 6.     | (a). Synchronous speed & speed of rotor for two and four- pole, 3- $\phi$ induction motor.<br>(b) Direction reversal by change of phase-sequence of 3-phase ac supply connections. | B    | A    | D    | C    |
| 7.     | Power factor improvement of induction motor using static capacitors.                                                                                                               | C    | B    | A    | D    |
| 8.     | Magnetization curve of a separately excited DC generator.                                                                                                                          | D    | C    | B    | A    |

**Lab Internal Exam:** ( 1 ONLY)

**Student Groups:**

**Course Coordinator:**

1604-\_\_-\_\_-\_\_

A:

B:

C:

D:

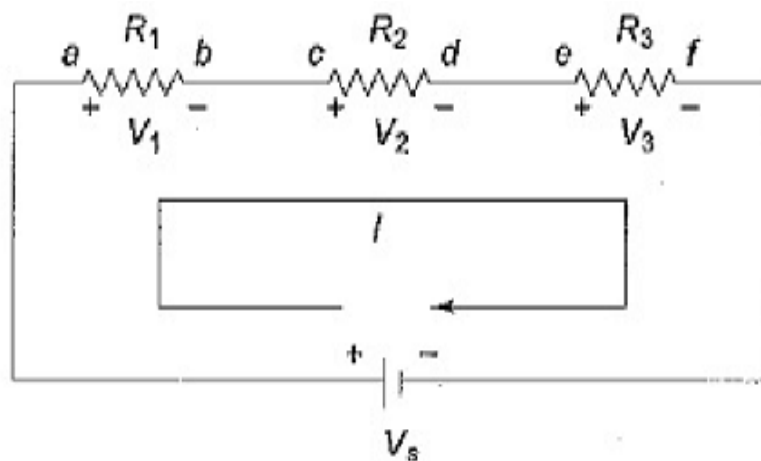
**EXPERIMENT 1a****VERIFICATION OF KVL****Aim:** To verify Kirchhoff's voltage law.**Apparatus Required:**

| Sl.No. | Apparatus        | Range     | Quantity |
|--------|------------------|-----------|----------|
| 1      | Trainer kit      |           | 1        |
| 2      | Resistance       |           | 3        |
| 3      | Voltmeter        | (0-30V)MC | 3        |
| 4      | Connecting wires | --        | Required |

**Theory:**

Kirchhoff's Voltage Law:

According to this law, in any closed circuit or mesh, the algebraic sum of EMFs acting in that circuit or mesh is equal to the voltage drops of each element of the circuit.

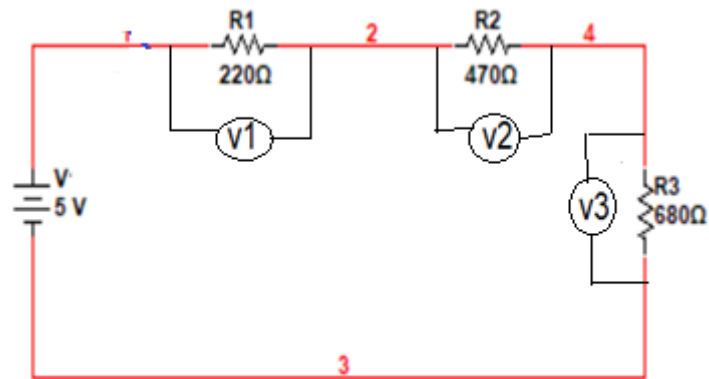


In mesh ABCD,  $V = I(R_1 + R_2 + R_3)$

**Procedure for KVL:**

1. Give the connections as per the circuit diagram.
2. Set a particular voltage value in RPS.
3. Note all the voltage reading
4. Repeat the same for different voltages

**Circuit for KVL:**



**Tabular Column for Verification of KVL:**

| Sl.No | Applied voltage<br>V in volts | V <sub>1</sub> in volts | V <sub>2</sub> in volts | V <sub>3</sub> in volts | V = V <sub>1</sub> +V <sub>2</sub> + V <sub>3</sub> |
|-------|-------------------------------|-------------------------|-------------------------|-------------------------|-----------------------------------------------------|
| 1     |                               |                         |                         |                         |                                                     |
| 2     |                               |                         |                         |                         |                                                     |
| 3     |                               |                         |                         |                         |                                                     |

**Result:**

KVL:  $V = V_1 + V_2 + V_3$   
 KVL is proved for the given circuits

**Discussion of Results:**

**EXPERIMENT 1b****VERIFICATION OF KCL****Aim:** To verify Kirchhoff's current law**Apparatus Required:**

| Sl.No. | Apparatus        | Range       | Quantity |
|--------|------------------|-------------|----------|
| 1      | Trainer Kit      |             | 1        |
| 2      | Resistance       |             | 3        |
| 3      | Ammeter          | (0-200mA)MC | 3        |
| 4      | Voltmeter        | (0-30V)MC   | 1        |
| 5      | Connecting wires | --          | Required |

**Theory:-**

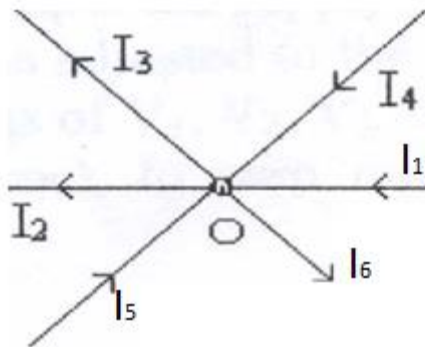
Kirchhoff's Current Law:

According to this law, in any network of wires carrying currents, the algebraic sum of all the currents meeting at a node is zero or the sum of all the incoming currents is equal to the sum of outgoing currents away from that node.

Let  $I_1, I_2, I_3, I_4, I_5, I_6$  be the currents at node O.  $I_1, I_4, I_5$  are the currents entering the node O and  $I_2, I_3, I_6$  are the currents leaving the node O. Then according to Kirchhoff's law,

$$I_1 - I_2 - I_3 + I_4 + I_5 - I_6 = 0$$

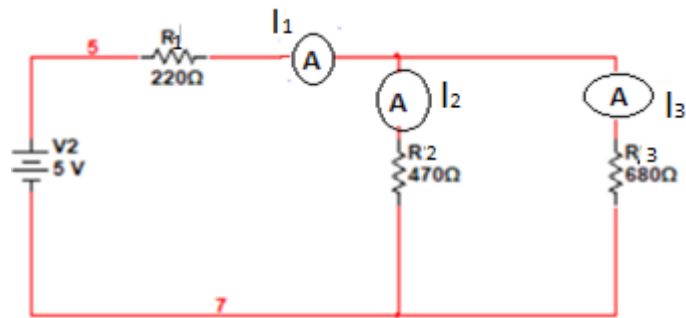
$$\text{i.e. } I_1 + I_4 + I_5 = I_2 + I_3 + I_6$$

**Procedure for KCL:**

1. Give the connections as per the circuit diagram.
2. Set a particular voltage value in RPS.
3. Note down the corresponding ammeter reading
4. Repeat the same for different voltages



**Circuit for KCL:**



**Tabular Column for Verification of KCL:**

| Sl.No | Applied voltage |               |                |                | $I_1 = I_2 + I_3$ (m Amps) |
|-------|-----------------|---------------|----------------|----------------|----------------------------|
|       | V in volts      | $I_1$ (mAmps) | $I_2$ (m Amps) | $I_3$ (m Amps) |                            |
| 1     |                 |               |                |                |                            |
| 2     |                 |               |                |                |                            |
| 3     |                 |               |                |                |                            |

**Precautions:**

1. Voltage control knob should be kept at minimum position.
2. Current control knob of RPS should be kept at maximum position.

**Result:**

KCL:  $I_1 = I_2 + I_3$

**Discussion of Results:**

**EXPERIMENT 1c****VERIFICATION OF SUPER POSITION THEOREM**

**Aim:** To verify Kirchhoff's current law, Kirchhoff's voltage law & super position theorem for the given circuit.

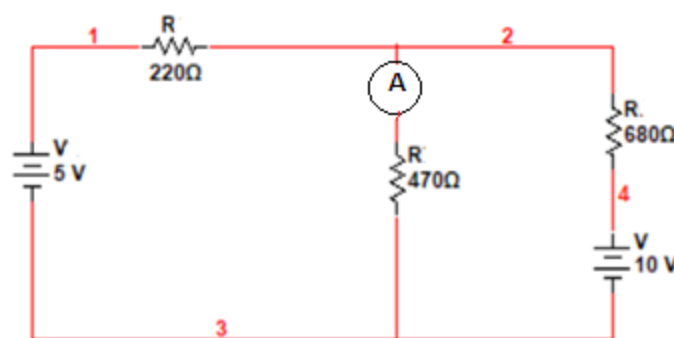
**Apparatus Required:**

| Sl.No. | Apparatus        | Range       | Quantity |
|--------|------------------|-------------|----------|
| 1      | Kit for theorem  |             | 1        |
| 2      | Ammeter          | (0-200mA)MC | 3        |
| 3      | Voltmeter        | (0-30V)MC   | 1        |
| 4      | Connecting wires | --          | Required |

**SUPERPOSITION THEOREM**

**Theory:-**

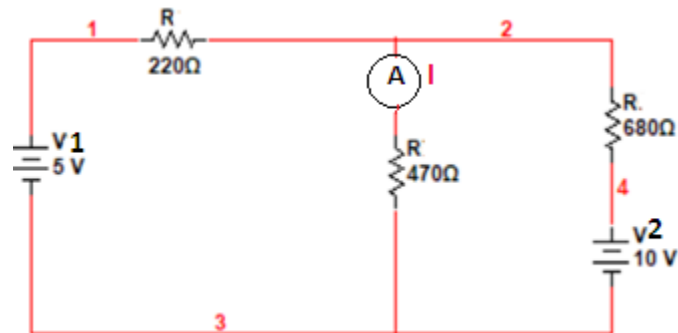
c) In a bilateral network consisting of a number of sources, the response in any branch is equal to sum of the responses due to individual sources taken one at a time with all other sources reduced to zero. When a network consists of several sources, this theorem helps us to find the current in any branch easily, considering only one source at a time.

**PROCEDURE:**

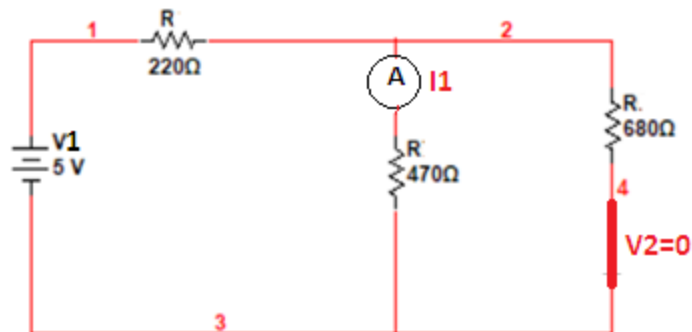
1. Connect the circuit as shown in FIG 1
2. Adjust the voltage of the source (1) to 5V and that of source (2) to 10V. Note the current (I) read by the ammeter.

3. Disconnect source (2) and short the terminals as in FIG 2 with source Voltage (1) at 5V read the ammeter current ( $I_1$ ).
4. Disconnect source and short the terminals as in FIG 3. With source (2) voltage at 10V read the ammeter current ( $I_2$ ).
5. Verify the equation  $I = I_1 + I_2$ .

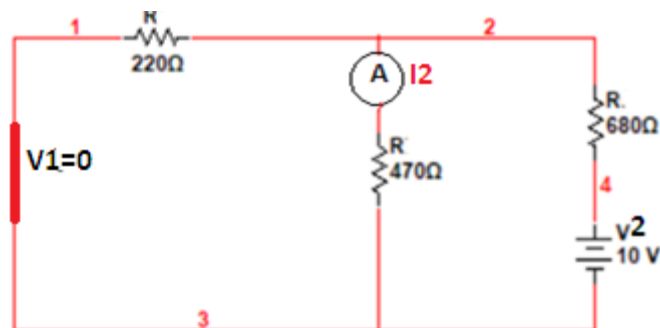
**CIRCUIT DIAGRAMS:**



**Fig 1: To find current in 470 ohms with both sources**



**Fig 2: To find current in 470 ohms with first source**



**Fig 3: To find current in 470 ohms with second source**

**OBSERVATION TABLE:**

| S.No | V <sub>1</sub> | V <sub>2</sub> | I(mA) | I <sub>1</sub> (mA) | I <sub>2</sub> (mA) | I=I <sub>1</sub> +I <sub>2</sub> (mA) |
|------|----------------|----------------|-------|---------------------|---------------------|---------------------------------------|
| 1    | 5              | 10             |       | --                  | --                  |                                       |
|      | 5              | 0              | ----  |                     | ----                |                                       |
|      | 0              | 10             | ---   | ----                |                     |                                       |

**Result:**

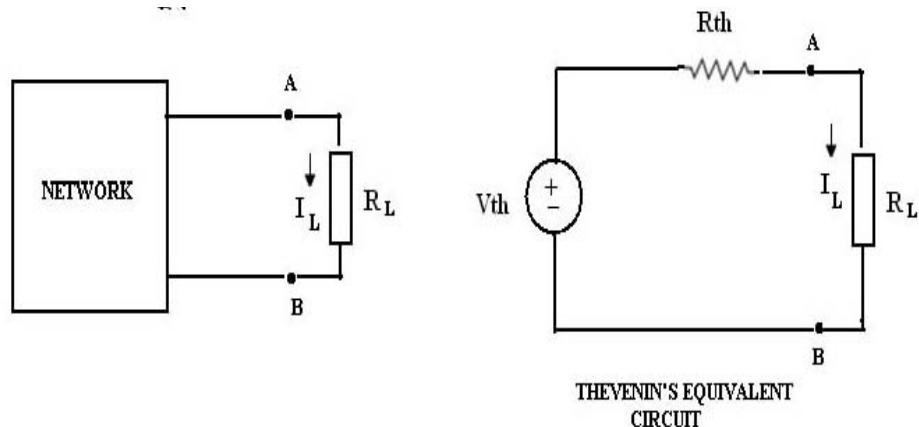
Superposition Theorem:  $I=I_1+I_2$

Superposition Theorem is proved for the given circuits

**Discussion of Results:**

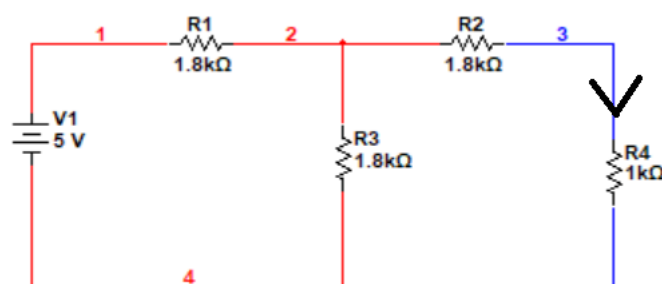
**EXPERIMENT 2a****VERIFICATION OF THEVENIN'S THEOREM****AIM:** To verify Thevenin's Theorem.**APPARATUS:**

| Sl.No. | Apparatus        | Range   | Quantity |
|--------|------------------|---------|----------|
| 1      | Trainer kit      |         | 1        |
| 2      | Ammeter          | (0-1A)  | 1        |
| 3      | Voltmeter        | (0-30V) | 1        |
| 4      | Connecting wires | --      | Required |

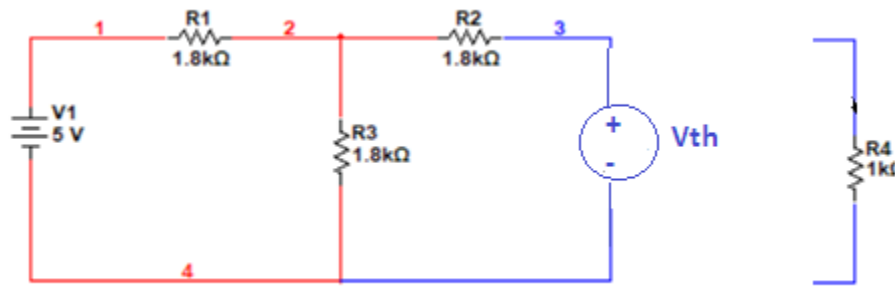
**THEORY:****THEVENIN'S THEOREM:**

Any linear bilateral network with respect to two terminals (A and B) can be replaced by a single voltage source  $V_{th}$  in series with a single resistance  $R_{th}$ . Where,  $V_{th}$  is the open circuit voltage across the load terminals and  $R_{th}$  is the internal resistance of the network as viewed back into the open circuited network from the terminals A and B with voltage sources and current sources replaced by their internal resistances. Then the current in the load resistance is given by,

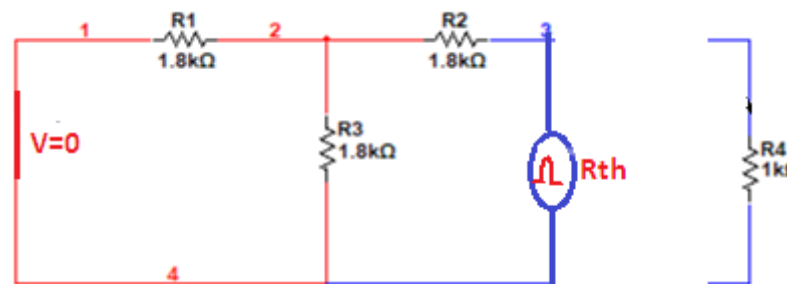
$$I_L = V_{th} / (R_{th} + R_L)$$



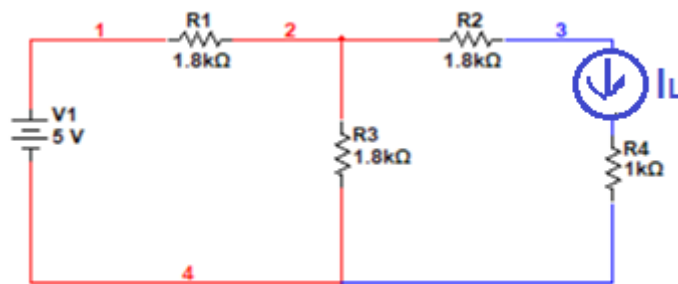
**CIRCUIT DIAGRAMS:**



**FIG 1: To find Thevenin's Voltage**



**FIG 2: To find Thevenin's resistance**



**FIG 3: To find load current**

**PROCEDURE:**

1. Connect the circuit as shown in FIG.1 and apply suitable voltage. Note down the open circuit voltage ( $V_{th}$ ).
2. Connect the circuit as shown in FIG.2 and note the Thevenin's resistance  $R_{th}$  by means of a multimeter.
3. Connect the circuit as shown in FIG.3. For a particular value of load resistance  $R_L$ , keeping the voltage of RPS at the same value as in step1, note the value of the current. Verify the current value obtained by applying the Thevenin's theorem i.e  $I_L$  should be equal to  $V_{th} / (R_{th} + R_L)$ .
4. Repeat step3 for various values of load resistances and compare with the calculated values, as obtained by applying Thevenin's theorem.

## BASIC ELECTRICAL ENGINEERING LAB, EED

5. Vary the input voltage and take three sets of readings (step 2 need not be repeated as long as the network is not changed).

### OBSERVATION TABLE:

$R_{th} (K\Omega) =$  \_\_\_\_\_

| <u>S.No.</u> | $V_s(V)$ | $V_{th}(V)$ | $R_L(K\Omega)$ | $I_L(\text{Measured Value in mA})$ | $I_L$ in mA (By applying theorem)<br>$I_L = V_{th}/(R_{th} + R_L)$ |
|--------------|----------|-------------|----------------|------------------------------------|--------------------------------------------------------------------|
| 1.           |          |             |                |                                    |                                                                    |
|              |          |             |                |                                    |                                                                    |
|              |          |             |                |                                    |                                                                    |

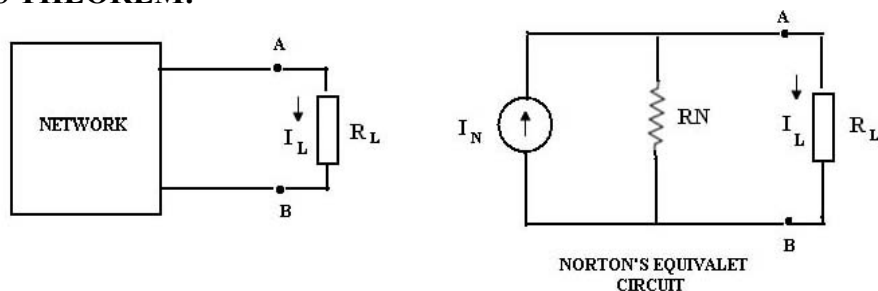
### RESULT:

**Load current by thevenins theorem :**  $I_L = V_{th}/(R_{th} + R_L)$

### DISCUSSION OF RESULT:

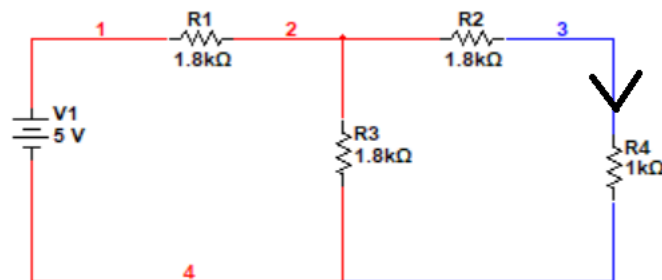
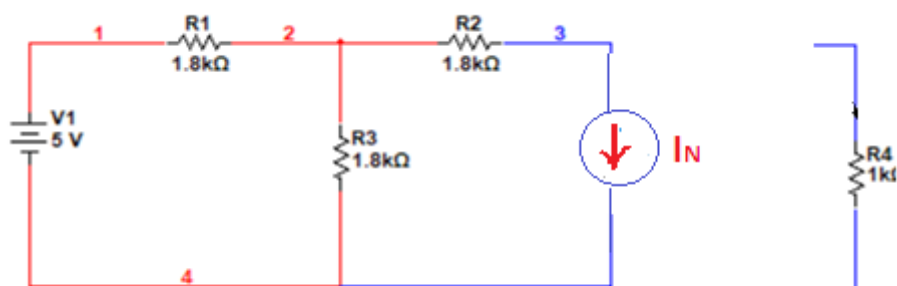
**EXPERIMENT 2b****VERIFICATION OF NORTON'S THEOREM****AIM:** To verify Norton's Theorem.**APPARATUS:**

| Sl.No. | Apparatus        | Range   | Quantity |
|--------|------------------|---------|----------|
| 1      | Trainer kit      |         | 1        |
| 2      | Ammeter          | (0-1A)  | 1        |
| 3      | Voltmeter        | (0-30V) | 1        |
| 4      | Connecting wires | --      | Required |

**THEORY:****NORTON'S THEOREM:**

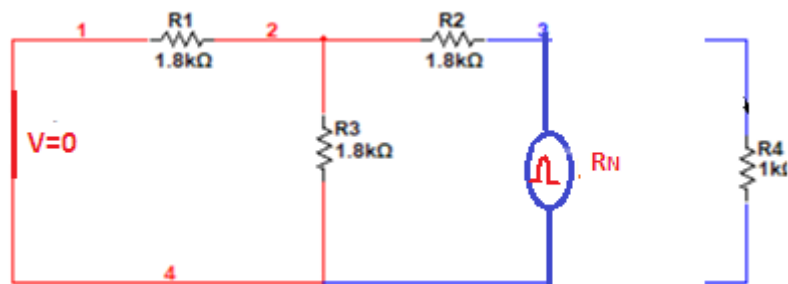
Any linear bilateral network with respect to a pair of terminals (A and B) can be replaced by a single current source  $I_N$  in parallel with a single resistance  $R_N$ . Where,  $I_N$  is the short circuit current in between the load terminals and  $R_N (= R_{th})$  is the internal resistance of the network as viewed back into the open circuited network from the terminals A and B with voltage sources and current sources replaced by their internal resistances. Then the current in the load resistance is given by,

$$I_L = I_N R_N / (R_N + R_L)$$

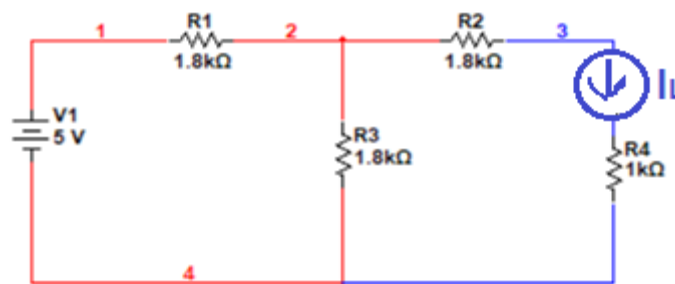
**CIRCUIT DIAGRAMS:**



**FIG 1: To find Norton's current**



**FIG 2: To find Norton's Resistance**



**FIG 3: To find load current**

**PROCEDURE:**

1. Connect the circuit as shown in FIG 1 and by applying suitable voltage through RPS, determine the short circuit current ( $I_N / I_{sc}$ ).
2. Connect the circuit as shown in FIG.2 and note the Norton's resistance  $R_N$  by means of a multimeter.
3. Connect the circuit as shown in FIG.3. For a particular value of load resistance  $R_L$ , keeping the voltage of RPS at the same value as in step1, note the value of the current. Verify the current value obtained by applying the Thevenin's theorem i.e  $I_L$  should be equal to  $V_{th} / (R_{th} + R_L)$ .
4. Repeat step3 for various values of load resistances and compare with the calculated values, as obtained by applying Thevenin's theorem.
5. Vary the input voltage and take three sets of readings (step 2 need not be repeated as long as the network is not changed)

**OBSERVATION TABLE:**

$$R_N(K\Omega) = \underline{\hspace{2cm}}$$

| S.No. | $V_s$<br>(V) | $I_N / I_{sc}$<br>(m Amps) | $R_L(K\Omega)$ | $I_L$ (Measured Value)<br>(m Amps) | $I_L$ (By applying theorem)<br>$I_L = I_N R_N / (R_N + R_L)$ (m A) |
|-------|--------------|----------------------------|----------------|------------------------------------|--------------------------------------------------------------------|
|       |              |                            |                |                                    |                                                                    |

|  |  |  |  |  |  |
|--|--|--|--|--|--|
|  |  |  |  |  |  |
|--|--|--|--|--|--|

**RESULT:**

**Load current by norton's theorem:**  $I_L = I_N R_N / (R_N + R_L)$

**DISCUSSION OF RESULT:**

**EXPERIMENT:3a****SINUSOIDAL STEADY STATE RESPONSE OF R-L SERIES CIRCUITS**

**Aim:** To verify voltage, current, power factor and impedance variations of R-L circuit to a sinusoidal input at steady state condition.

**Apparatus Required:**

| S.No | Name of Apparatus | Range/Rating | Required |  |
|------|-------------------|--------------|----------|--|
| 1    | Trainer Kit       |              | 1        |  |
| 2    | Voltmeter         | 0-10V        | 2        |  |
| 3    | Ammeter           | 0-20mA       | 1        |  |
| 4    | Connecting Wires  |              | APR      |  |

**THEORY:-****RL Series Circuit:**

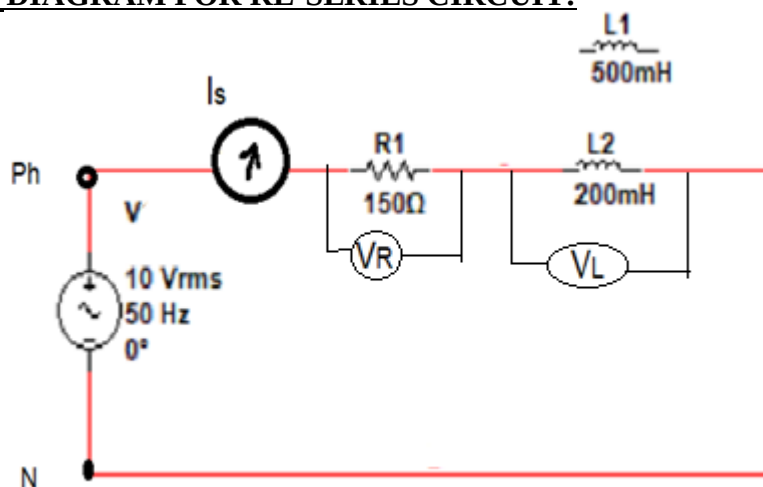
Any circuit behavior depends on the parameters of the elements available in it. Such elements, which drastically changes the parameters of the circuit, are resistance, inductance and capacitance.

In series RL network when a voltage  $V_s$  is applied then the current  $I_s$  flows into the circuit which creates a voltage drops across resistance  $V_R$  and inductance  $V_L$ . The applied source voltage  $V$  leads  $I_s$  by an angle  $\theta$ .

**Procedure:**

1. Connect circuit as shown in the circuit diagram
2. Apply the voltage using Function Generator and set the frequency to 50Hz
3. Note all the readings and tabulate them in the given tabular form.
4. Repeat the procedure with different R&L values

Circuit diagram of RL series circuit:

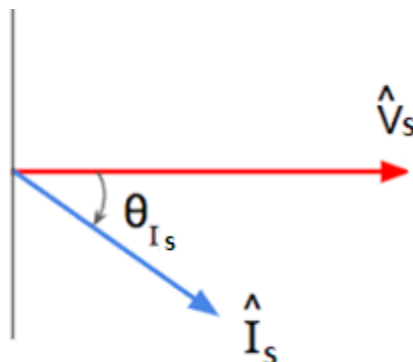
**CIRCUIT DIAGRAM FOR RL-SERIES CIRCUIT:**

**Observation Table for series RL circuit:****PRACTICAL CALCULATIONS:**

| $V_s$ | $R(\Omega)$ | $L$ (mH) | $I_s(\text{mA})$ | $V_R(\text{V})$ | $V_L(\text{V})$ | $Z = V_s/I_s(\Omega)$ | $\theta = \tan^{-1}\left(\frac{V_L}{V_R}\right)$<br>(Degree) | Power Factor<br>$\cos \theta$<br>(No Units) |
|-------|-------------|----------|------------------|-----------------|-----------------|-----------------------|--------------------------------------------------------------|---------------------------------------------|
| 10.48 | 150         | 200      |                  |                 |                 |                       |                                                              |                                             |
| 10.48 | 150         | 500      |                  |                 |                 |                       |                                                              |                                             |

**THEORETICAL CALCULATIONS:**

| $V_s$ | $R(\Omega)$ | $L(\text{mH})$ | $I_s = \frac{V_s}{ Z }(\text{mA})$ | $V_R = I_s R$<br>(V) | $V_L = I_s X_L$<br>(V) | $Z = R + jX_L$<br>( $\Omega$ ) | $\theta = \tan^{-1}\left(\frac{X_L}{R}\right)$<br>(Degree) | Power Factor<br>$\cos \theta$<br>(No Units) |
|-------|-------------|----------------|------------------------------------|----------------------|------------------------|--------------------------------|------------------------------------------------------------|---------------------------------------------|
| 10.48 | 150         | 200            |                                    |                      |                        |                                |                                                            |                                             |
| 10.48 | 150         | 500            |                                    |                      |                        |                                |                                                            |                                             |

**VECTOR DIAGRAM:**

$$\theta_z = \theta_V - \theta_i$$

Since angle of voltage  $\theta_V = 0$ ,  $\theta_i = \theta_z$

**RESULT:**

**RL-Series circuit:** The applied source voltage V leads  $I_s$  by an angle  $\theta =$  \_\_\_\_\_ for  
R = \_\_\_\_\_, L = \_\_\_\_\_.

**DISCUSSION OF RESULT:**

**EXPERIMENT:3b****SINUSOIDAL STEADY STATE RESPONSE OF R-C SERIES CIRCUITS**

**Aim:** To verify voltage, current, power factor and impedance variations of R-L and R-C circuits to a sinusoidal input at steady state condition.

**Apparatus Required:**

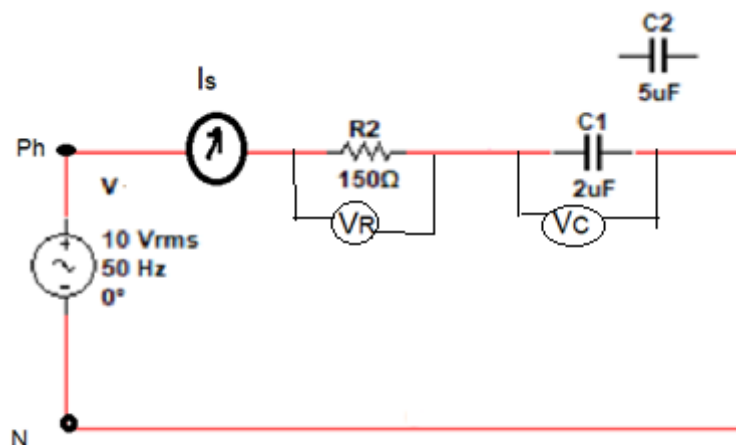
| S.No | Name of Apparatus | Range/Rating | Required |  |
|------|-------------------|--------------|----------|--|
| 1    | Trainer Kit       |              | 1        |  |
| 2    | Voltmeter         | 0-10V        | 2        |  |
| 3    | Ammeter           | 0-20mA       | 1        |  |
| 4    | Connecting Wires  |              | APR      |  |

**THEORY:-****RC Series Circuit:**

In series RC network when a voltage  $V_s$  is applied then the current  $I_s$  flows into the circuit which creates a voltage drops across resistance  $V_R$  and capacitance  $V_C$ . The applied source voltage  $V$  lags behind  $I_s$  by an angle  $\theta$ .

**Procedure:****For series RC circuit**

1. Connect circuit as shown in the circuit diagram
2. Apply the voltage using Function Generator and set the frequency to 50Hz
3. Note all the readings and tabulate them in the given tabular form.
4. Repeat the procedure with different R&C values

**CIRCUIT DIAGRAM FOR RC-SERIES CIRCUIT:**

**OBSERVATION TABLE:**  
**RC-SERIES CIRCUIT:**  
**PRACTICAL CALCULATIONS:**

| $V_s$ | $R(\Omega)$ | $C(\mu F)$ | $I_s(mA)$ | $V_R(V)$ | $V_C(V)$ | $Z = V_s/I_s(\Omega)$ | $\theta = \tan^{-1}\left(\frac{V_C}{V_R}\right)$<br>(Degree) | Power Factor $\cos \theta$<br>(No Units) |
|-------|-------------|------------|-----------|----------|----------|-----------------------|--------------------------------------------------------------|------------------------------------------|
| 10.48 | 150         | 2          |           |          |          |                       |                                                              |                                          |
| 10.48 | 150         | 5          |           |          |          |                       |                                                              |                                          |

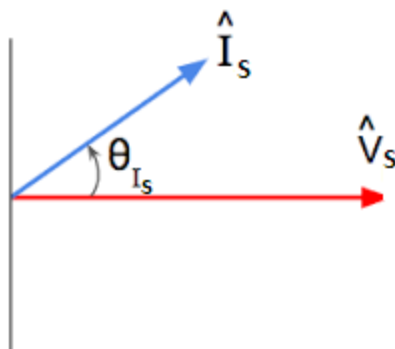


**THEORETICAL CALCULATIONS:**

| $V_s$ | $R(\Omega)$ | $C(\mu F)$ | $I_s = \frac{V_s}{ Z }$<br>(mA) | $V_R = I_s R$<br>(V) | $V_C = I_s X_C$<br>(V) | $Z = R - jX_C$<br>( $\Omega$ ) | $\theta = \tan^{-1}\left(\frac{-X_C}{R}\right)$<br>(Degree) | Power Factor $\cos \theta$<br>(No Units) |
|-------|-------------|------------|---------------------------------|----------------------|------------------------|--------------------------------|-------------------------------------------------------------|------------------------------------------|
| 10.48 | 150         | 2          |                                 |                      |                        |                                |                                                             |                                          |
| 10.48 | 150         | 5          |                                 |                      |                        |                                |                                                             |                                          |



**VECTOR DIAGRAM:**



**RESULT:**

**RC-Series circuit:** The applied source voltage  $V$  lags  $I_s$  by an angle  $\theta = \underline{\hspace{2cm}}$  for

BASIC ELECTRICAL ENGINEERING LAB, EED

R = \_\_\_\_\_, C = \_\_\_\_\_.

**DISCUSSION OF RESULT:**



**EXPERIMENT: 4a****MEASUREMENT OF PHASE VOLTAGE/CURRENT, LINE VOLTAGE/CURRENT AND POWER IN BALANCED THREE-PHASE STAR CONNECTED CIRCUIT.**

**Aim :** To study the balanced three phase system for star connected load.

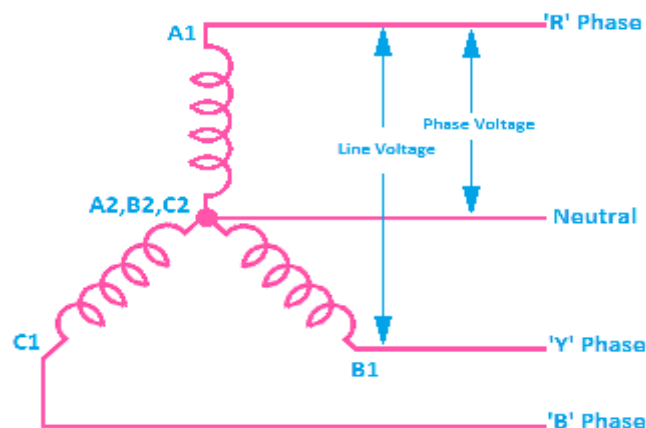
**Apparatus Required:**

| S.No | Name of Apparatus  | Range/Rating       | Required |
|------|--------------------|--------------------|----------|
| 1    | Three phase Variac | 440/0-440V         | 1        |
| 2    | Ammeter (AC)       | 10A                | 1        |
| 3    | Voltmeter (AC)     | 600V               | 1        |
| 4    | Rheostats          | 25ohms/5A          | 3        |
| 5    | Wattmeter Meter    | 0-300V/5A UPF      | 1        |
| 6    | Connecting wires   | As per requirement |          |

**THEORY:- Star Connection**

Any three phase system, either supply system or load can be connected in two ways either star or delta. **In** this connection, the terminating ends of all winding are connected together called as neutral point & starting ends are connected to their individual phases as shown below.

- In this experiment we are using balanced load.
- The load is said to be balanced when
  - Voltages across three phases are equal & phases are displaced by  $120^\circ$  electrical.
  - The impedance of each phase of load is same.
  - The resulting current in all the three phases are equal & displaced by  $120^\circ$  electrical from each other
  - Active power & reactive volt amperes of each is equal.

**Star Connection****Some term related to 3 phase system**

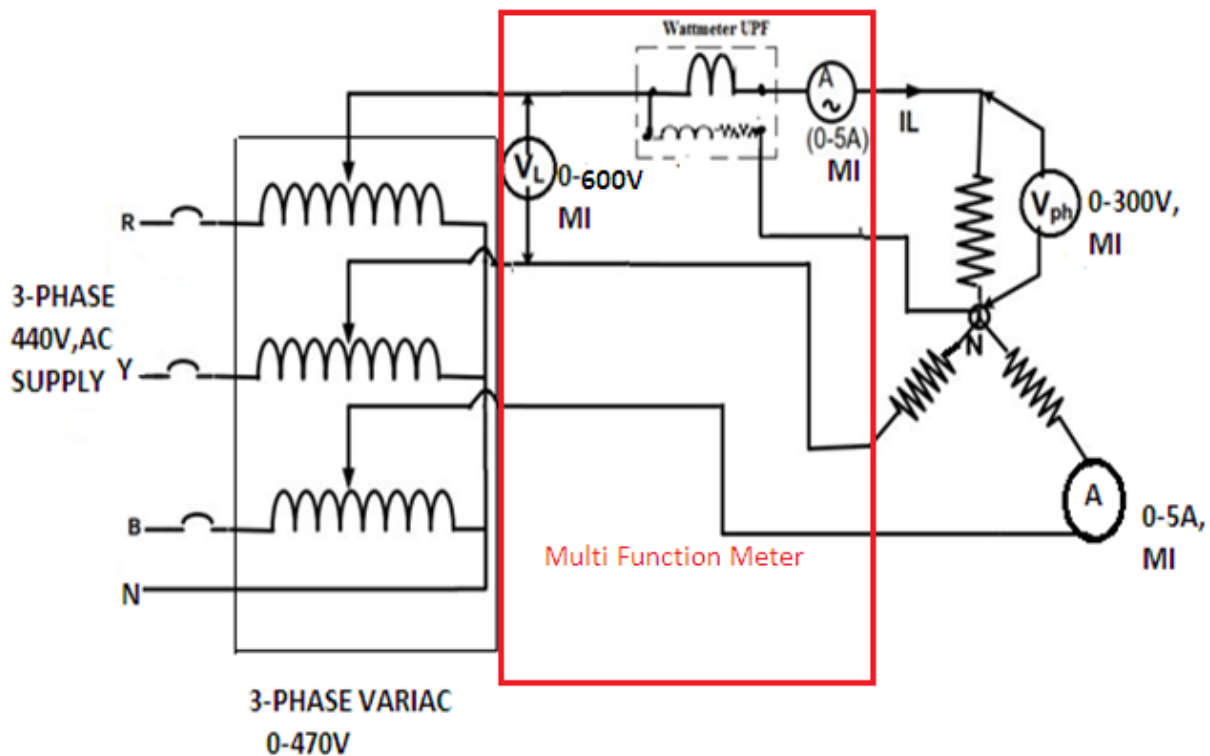
- i. **Line Voltage** - The voltage between any two line of 3 ph load is called as line voltage e.g.  $V_{RY}$ ,  $V_{YB}$  &  $V_{BR}$ .. For balance system all are equal in magnitude.

ii. **Line Current**– The current in each line is called as line current e.g.  $I_R$ ,  $I_Y$ , &  $I_B$ . They are equal in magnitude for balance system.

iii. **Phase Voltage**– The voltage across any phase of three phase load is called as phase voltage.  $V_{RN}$ ,  $V_{YN}$ , &  $V_{BN}$  are phase voltage

iv. **Phase Current**– current passing through each phase of load is called as phase current.

**CIRCUIT DIAGRAM:**



**OBSERVATION TABLE:**

| S.No | Line Voltage( $V_L$ ) (V) | Phase Voltage( $V_{Ph}$ ) (V) | Phase Current( $I_{Ph}$ ) (A) | Line Current( $I_L$ ) (A) | 3-Phase Power Measured (KW) | 3-Phase Power Calculated $W = \sqrt{3} V_L I_L \cos \phi$ (KW) |
|------|---------------------------|-------------------------------|-------------------------------|---------------------------|-----------------------------|----------------------------------------------------------------|
|      |                           |                               |                               |                           |                             |                                                                |
|      |                           |                               |                               |                           |                             |                                                                |

**PROCEDURE:**

1. Connect circuit as shown in the circuit diagram
2. Set 3-ph variac to minimum position.
3. Switch on the main supply

## BASIC ELECTRICAL ENGINEERING LAB, EED

4. Apply a voltage by using 3-ph variac
5. Note the readings of ammeter, voltmeter & wattmeter meter.
6. Repeat the above procedure by changing supply voltage.

### RESULT:

In star connected load it is observed that the relation between phase & Line quantities are:

**Voltage:** Line voltage ( $V_L$ ) =  $\sqrt{3}$  phase voltage ( $V_{ph}$ )

**Current:** Line current ( $I_L$ ) = Phase current ( $I_{ph}$ )

**3-phase Power:**  $\sqrt{3} V_L I_L \cos\phi = 3 V_{ph} I_{ph} \cos\phi$

Where  $\phi$  is phase angle & it depends on type of load i.e. inductive, capacitive or resistive.

**LOAD USED is RESISTIVE ,Therefore the power factor  $\cos\phi=1$**

### DISCUSSION OF RESULT:

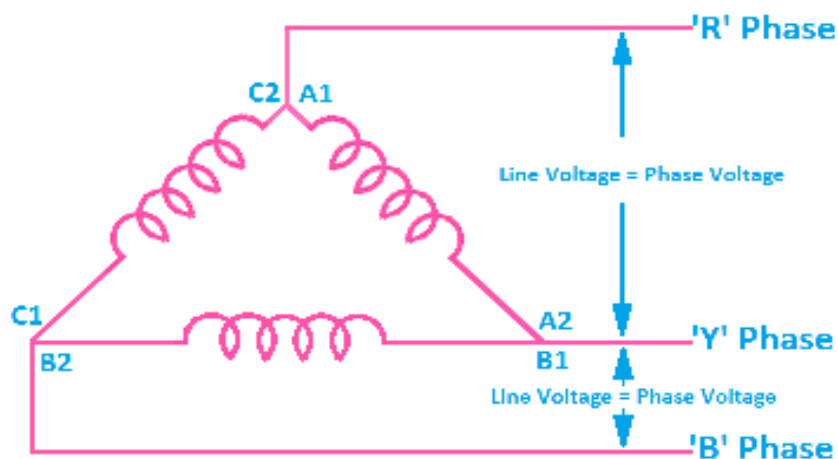
**EXPERIMENT: 4b****MEASUREMENT OF PHASE VOLTAGE/CURRENT, LINE VOLTAGE/CURRENT AND POWER IN BALANCED THREE-PHASE DELTA CONNECTED CIRCUIT.****Aim :** To study the balanced three phase system for star connected load.**Apparatus Required:**

| S.No | Name of Apparatus  | Range/Rating       | Quantity |
|------|--------------------|--------------------|----------|
| 1    | Three phase Variac | 440/0-440V         | 1        |
| 2    | Ammeter (AC)       | 10A                | 1        |
| 3    | Voltmeter (AC)     | 600V               | 1        |
| 4    | Rheostats          | 25ohms/5A          | 3        |
| 5    | Wattmeter Meter    | 0-300V/5A UPF      | 1        |
| 6    | Connecting wires   | As per requirement |          |

**THEORY:- Delta Connection**

Any three phase system, either supply system or load can be connected in two ways either star or delta. If the terminating end of one winding is connected to starting end of other & if connection are continued for all the three windings in this fashion we get closed loop. The three supply lines are taken out from three junctions. This is called as three phase delta connected system.

- In this experiment we are using balanced load.
- The load is said to be balanced when
  - Voltages across three phases are equal & phases are displaced by  $120^\circ$  electrical.
  - The impedance of each phase of load is same.
  - The resulting current in all the three phases are equal & displaced by  $120^\circ$  electrical from each other
  - Active power & reactive volt amperes of each is equal.

**DELTA CONNECTION**

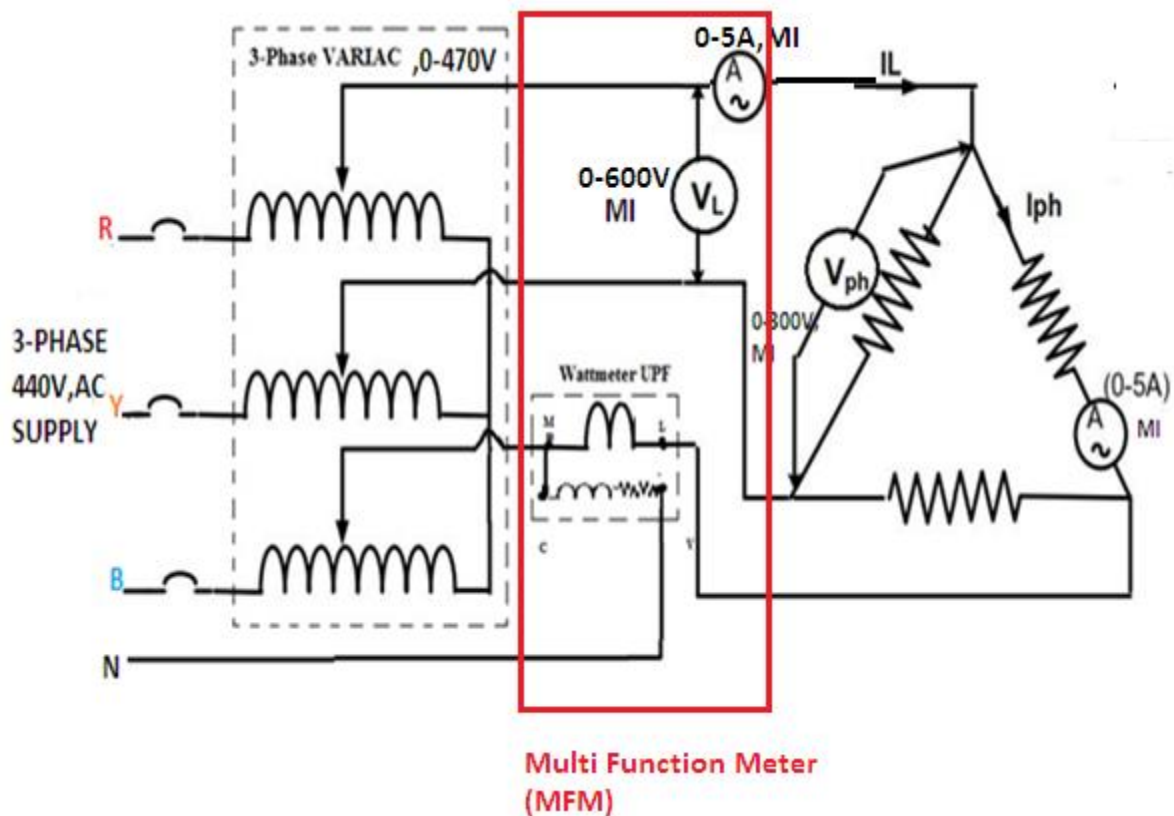
Some term related to 3 phase system

- Line Voltage - The voltage between any two line of 3 ph load is called as line voltage e.g. VRY, VYB & VBR.. For balance system all are equal in magnitude.

## BASIC ELECTRICAL ENGINEERING LAB, EED

- ii. Line Current– The current in each line is called as line current e.g.  $I_R$ ,  $I_Y$ , &  $I_B$ . They are equal in magnitude for balance system.
- iii. Phase Voltage– The voltage across any phase of three phase load is called as phase voltage.  $V_{RN}$ ,  $V_{YN}$ , &  $V_{BN}$  are phase voltage
- iv. Phase Current– current passing through each phase of load is called as phase current.

### CIRCUIT DIAGRAM:



### OBSERVATION TABLE:

| S.No | Line Voltage( $V_L$ ) (V) | Phase Voltage( $V_{Ph}$ ) (V) | Phase Current( $I_{Ph}$ ) (A) | Line Current( $I_L$ ) (A) | 3-Phase Power Measured (KW) | 3-Phase Power Calculated $W = \sqrt{3} V_L I_L \cos \phi$ (KW) |
|------|---------------------------|-------------------------------|-------------------------------|---------------------------|-----------------------------|----------------------------------------------------------------|
|      |                           |                               |                               |                           |                             |                                                                |
|      |                           |                               |                               |                           |                             |                                                                |
|      |                           |                               |                               |                           |                             |                                                                |

### PROCEDURE:

1. Connect circuit as shown in the circuit diagram
2. Set 3-ph variac to minimum position.
3. Switch on the main supply
4. Apply a voltage by using 3-ph variac
5. Note the readings of ammeter, voltmeter & wattmeter meter.

6. Repeat the above procedure by changing supply voltage.

### RESULT

In delta connected load it is observed that the relation between phase & Line quantities are:

Voltage: Line voltage ( $V_L$ ) = phase voltage ( $V_{ph}$ )

Current: Line current ( $I_L$ ) =  $\sqrt{3}$  Phase current ( $I_{ph}$ )

3-phase Power:  $\sqrt{3} V_L I_L \cos\phi = 3 V_{ph} I_{ph} \cos\phi$

Where  $\phi$  is phase angle & it depends on type of load i.e. inductive, capacitive or resistive.

**LOAD USED is RESISTIVE , Therefore the power factor  $\cos\phi=1$**

### DISCUSSION OF RESULT:

**EXPERIMENT:5****LOADING OF SINGLE PHASE TRANSFORMER**

**AIM:** To measure the primary and secondary: current, voltage and power of the given transformer.

**NAME PLATE DETAILS:** **P=3KVA**, **LV:HV=115V:230V**,  **$I_{LV}:I_{HV}$  = 26.08A:13.04A**

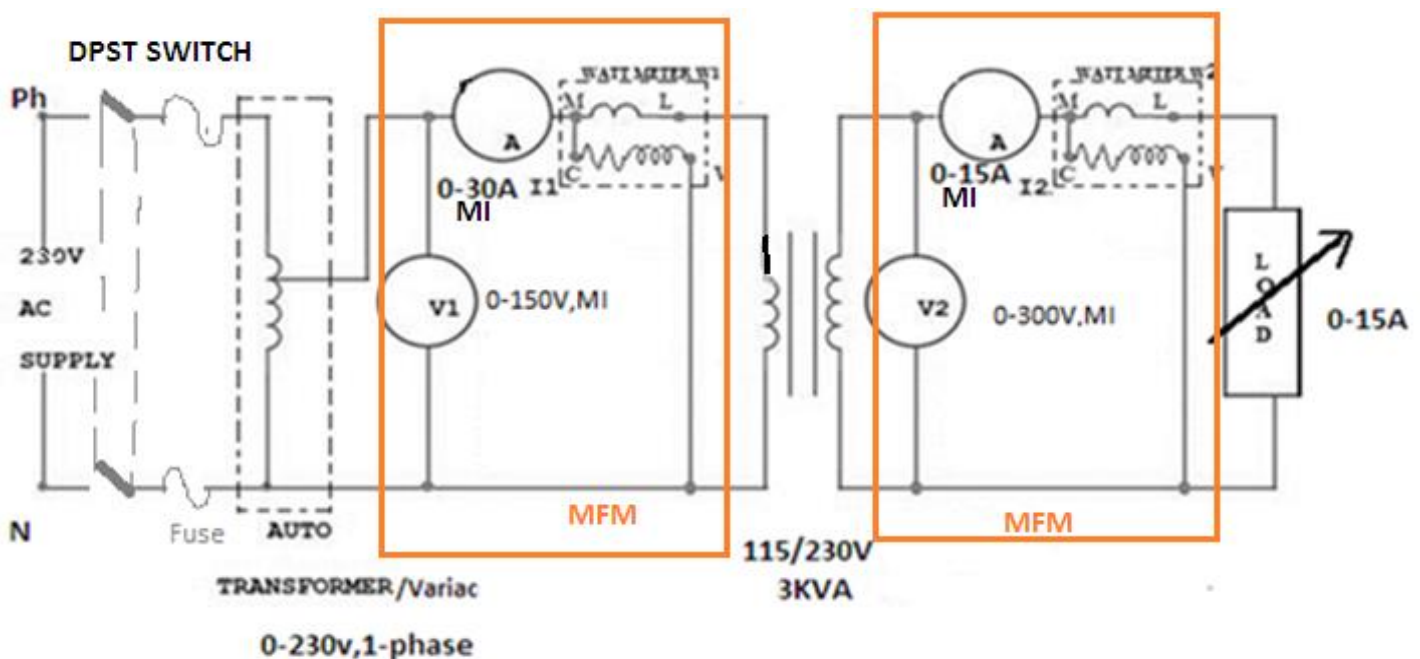
**APPARATUS:**

| Sl.No. | Apparatus                | Range            | Quantity |
|--------|--------------------------|------------------|----------|
| 1      | Single phase Transformer | 115/230V         | 1        |
| 2      | Auto transformer         | 230/(0-270) V    | 1        |
| 3      | Ammeter(MI)              | (0-10/20A)       | 2        |
| 4      | Voltmeter(MI)            | (0-150/300V)     | 2        |
| 5      | Resistive Load Box       | 0-10KW           | 1        |
| 6      | Wattmeter                | (0-300V/20A) UPF | 2        |
| 7      | Connecting wires         | --               | APR      |

**THEORY:**

Transformer works on the principal of mutual induction, which implies-when an ac voltage is applied to the primary coil, then the ac current flows in the primary coil giving rise to alternating flux. This alternating flux induces emf in the primary coil, called self induced emf and links with the secondary coil thus inducing an emf in the secondary coil, called mutually induced emf.

**CIRCUIT DIAGRAM:**



**PROCEDURE:**

1. Connect the circuit as shown in figure above
2. Apply the rated voltage of primary winding(115V) using auto transformer
3. Record the readings of ammeter (I1), voltmeter(V1) and wattmeter (W1) connected in primary side
4. Record the readings of ammeter (I2), voltmeter(V2) and wattmeter (W2) connected in secondary side
5. Increase the load in steps from no-load(I2=0 to full load I2=13.04A) by using resistive load box
6. Tabulate the readings for every change in load

**OBSERVATIONS:**

| S.No. | V <sub>1</sub> (V)<br>Primary Voltage | I <sub>1</sub> (A)<br>Primary Current | W <sub>1</sub> (Watts)<br>Primary power | V <sub>2</sub> (V)<br>Secondary voltage | I <sub>2</sub> (A)<br>Secondary current | W <sub>2</sub> (Watts)<br>Secondary power |
|-------|---------------------------------------|---------------------------------------|-----------------------------------------|-----------------------------------------|-----------------------------------------|-------------------------------------------|
| 1     | 115V                                  |                                       |                                         |                                         |                                         |                                           |
| 2     | 115V                                  |                                       |                                         |                                         |                                         |                                           |
| 3     | 115V                                  |                                       |                                         |                                         |                                         |                                           |
| 4     | 115V                                  |                                       |                                         |                                         |                                         |                                           |

**RESULT:**

**DISCUSSION OF RESULT:** As the load connected in secondary is resistive (power factor  $\cos\theta=1$ ).

Therefore, wattmeter W2 reads  $W_2 = V_2 \cdot I_2 \cdot \overset{=1}{\cancel{\cos\theta}}$

$$\Rightarrow W_2 = V_2 \cdot I_2$$



**EXPERIMENT:6****(a) SYNCHRONOUS SPEED & SPEED OF ROTOR FOR TWO AND FOUR- POLE, 3- $\phi$  INDUCTION MOTOR.****(b) DIRECTION REVERSAL BY CHANGE OF PHASE- SEQUENCE OF 3- PHASE AC SUPPLY CONNECTIONS.****AIM:** To verify the speed of 3 $\Phi$  Induction motor for:

- a) 2 pole Configuration circuit & 4pole Configuration circuit
- b) To change the direction of a three phase induction motor rotation.

**NAME PLATE DETAILS:**

3-Phase Induction Motor (2-Pole)

Voltage: -----

Power: -----

3-Phase Induction Motor (4-Pole)

Voltage: -----

Power: -----

Current: -----

Speed (Nr): -----

Current: -----

Speed (Nr): -----

**APPARATUS REQUIRED:**

| SL.NO | NAME OF EQUIPMENT | TYPE    | RANGE    | QUANTITY |
|-------|-------------------|---------|----------|----------|
| 1     | Tachometer        | Digital | 0-99999  | 1        |
| 2     | 3 $\Phi$ variac   |         | (0-470V) | 1        |

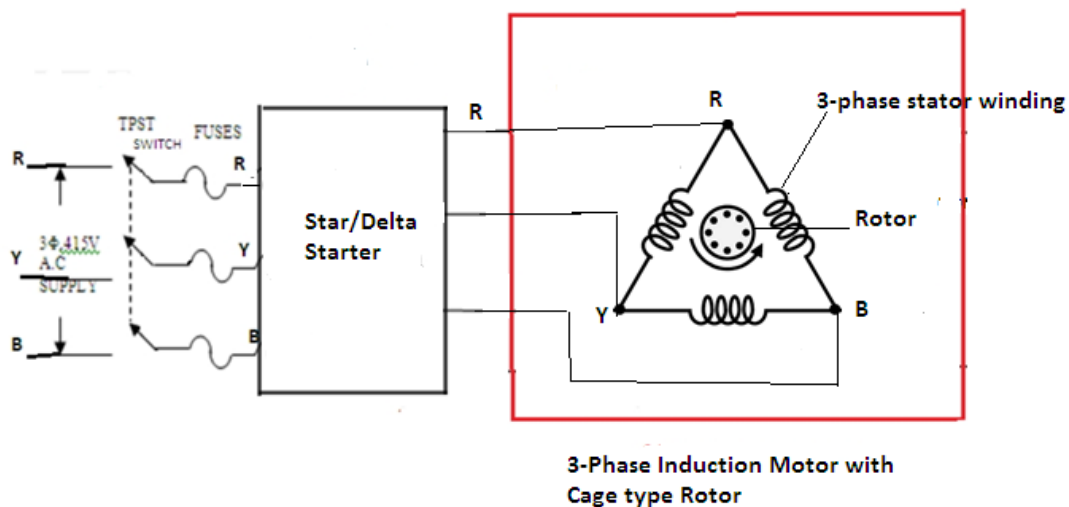
**Theory**

The speed with which stator field revolves is the synchronous speed. It is the speed at which the rotating magnetic field (mmf) rotates. if a three phase induction motor is having 6 poles with 50Hz supply frequency then the synchronous speed will be ( $N_s = 120f/p$ ) 1000rpm. Similarly 4 pole machines must work on 1500RPM where as 2 Pole machine at 3000RPM.

In order for an induction motor to make **torque**, there must be at least some difference between the stator field (synchronous) speed and the rotor speed. That difference is called “slip.” It is why, when you look at a motor name plate, the motor rated speed will always be less than synchronous speed. The direction of rotation of a 3 phase induction motor can be reversed by interchanging any two of the three motor supply lines.

Let the phase sequence of the three-phase voltage applied to the stator winding is R-Y-B. If this sequence is changed to R-B-Y, it is observed that direction of rotation of the field is reversed i.e., the field rotates counter clockwise rather than clockwise. However, the number of poles and the speed at which the magnetic field rotates remain unchanged. Thus it is necessary only to change the phase sequence in order to change the direction of rotation of the magnetic field. For a three-phase supply, this can be done by interchanging any two of the three lines

**CIRCUIT DIAGRAM:**



### PROCEDURE FOR (a) POLE CHANGE:

- 1) Make the connections as per the circuit diagram
- 2) Switch on the supply and apply the rated voltage to motor by using 3 phase variac
- 3) With the help of tachometer note the speed with different pole configuration, also observe the direction of rotation of Rotor.
- 4) Bring back the variac to initial zero position and switch off the supply
- 5) Calculate the slip by using given formulas

**Synchronous Speed( $N_s$ )= $120f/p$**

For 2-pole:  $N_s=(120*50)/2=3000\text{rpm}$

For 4-pole:  $N_s=(120*50)/4=1500\text{rpm}$

### OBSERVATIONS:

| S.NO | Poles | Synchronous Speed ( $N_s$ ) R.P.M | Rotor Speed ( $N_r$ ) R.P.M |     | Slip (%) = $[(N_s-N_r)/N_s]*100$ |     |
|------|-------|-----------------------------------|-----------------------------|-----|----------------------------------|-----|
|      |       |                                   | RYB                         | RBV | RYB                              | RBV |
| 1    | 2     |                                   |                             |     |                                  |     |
| 2    | 4     |                                   |                             |     |                                  |     |

### PROCEDURE FOR (b) DIRECTION REVERING

- 1) Make the connections by interchanging the 3-phase Ac supply terminals from R-Y-B to R-B-Y in the circuit diagram
- 2) Switch on the supply and apply the rated voltage to motor by using 3 phase variac
- 3). Speed of the motor is measured by a speedometer and direction of the rotation is observed (It will be opposite to that of case 1(a)).
- 4). TPST switch is made OFF (OPEN), thereby disconnecting 3-phase Induction Motor from supply.

**RESULT:** The speed of 3 $\Phi$  Induction motor for 2 pole & 4pole Configuration is measured

## BASIC ELECTRICAL ENGINEERING LAB, EED

and the direction of rotation with change in phase sequence of AC supply for a three phase induction motor is observed.

### **DISCUSSION OF RESULT:**

(a) Synchronous speed ( $N_s$ ) is inversely proportional to No. of Poles

For 2-pole:  $N_s = (120 \times 50) / 2 = 3000 \text{ rpm}$

For 4-pole:  $N_s = (120 \times 50) / 4 = 1500 \text{ rpm}$

(b) R-Y-B phase sequence of AC supply,  $N_r = \text{-----rpm}$  (direction-clockwise)

R-B-Y phase sequence of AC supply,  $N_r = \text{-----rpm}$  (direction- Anticlockwise)

### **EXPERIMENT-7**

### **POWER FACTOR IMPROVEMENT OF INDUCTION MOTOR USING STATIC CAPACITORS**

**AIM: To verify Power factor improvement of Induction Motor using static capacitors**

**NAME PLATE DETAILS:**

Rated Voltage: \_\_\_\_\_

Rated Current : \_\_\_\_\_

Rated Power : \_\_\_\_\_

Rated Speed: \_\_\_\_\_

**APPARATUS:**

| S.No | Description             | Range     | Type   | Quantity           |
|------|-------------------------|-----------|--------|--------------------|
| 1    | 1 Phase Induction Motor | 1Hp       |        | 1                  |
| 2    | Ammeter                 | 0-10A/20A | MI     | 1                  |
| 3    | Voltmeter               | 0-300     | MI     | 1                  |
| 4    | Wattmeter               | 0-300/10A | Dynamo | 1                  |
| 5    | Connecting wires        |           |        | As per Requirement |
| 6    | Capacitor Bank          |           |        | 1                  |

**THEORY:**

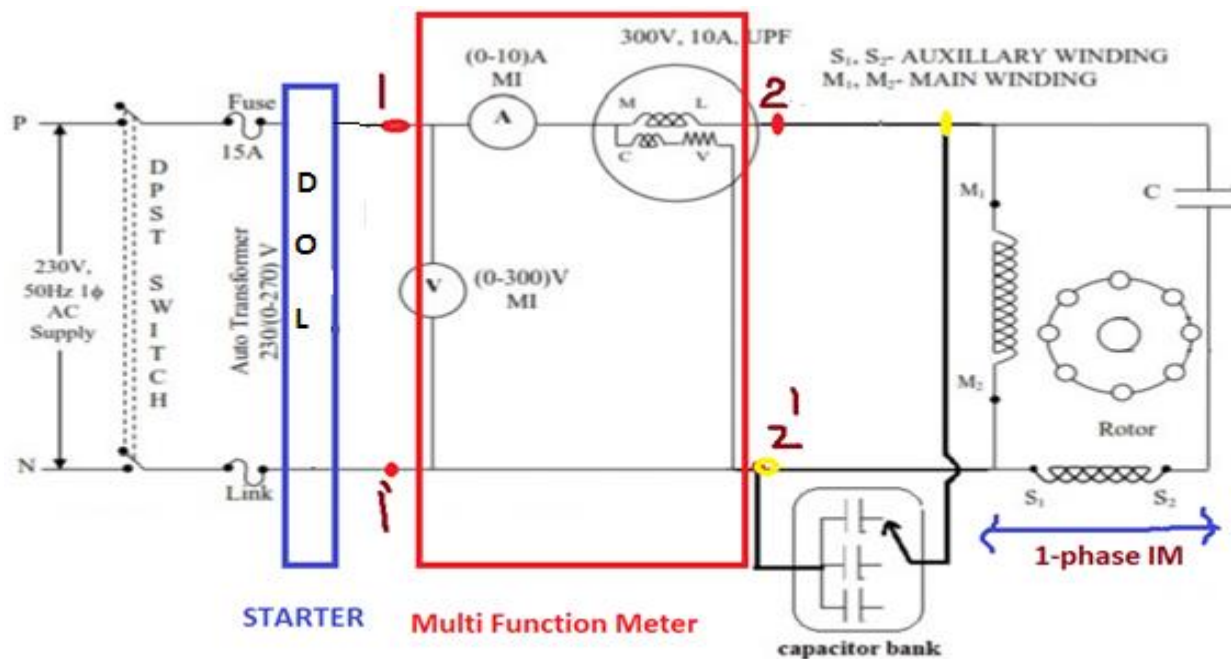
Low power factor is undesirable from economic point of view. Normally, the power factor of the load on the supply system is lower than 0.8. The following are the causes of low power factor:

- (i) Most of the a.c. motors are of induction type (1 $\phi$  and 3 $\phi$  induction motors) which have low lagging power factor.
- (ii) Arc & electric discharge lamps & industrial heating furnaces operate at low lagging PF.
- (iii) The load on the power system is varying ; being high during morning and evening and low at other times. During low load period, supply voltage is increased which increases the magnetization current. This results in the decreased power factor.

Static capacitor. The power factor can be improved by connecting capacitors in parallel with the equipment operating at lagging power factor. The capacitor (generally known as static) draws a leading current and partly or completely neutralize the lagging reactive component

of load current. This raises the power factor of the load. For three-phase loads, the capacitors can be connected in delta or star as shown in Fig. 6.4. Static capacitors are invariably used for power factor improvement in factories.

**CIRCUIT DIAGRAM:**

**Procedure :**

1. Connections are made as per circuit diagram
2. Keep the starter in off position and capacitor bank switch in Zero position & switch on the supply by closing the DPST switch.
3. Switch on the starter & Take down all the meter readings by setting auto transformer to 230V
4. Switch the capacitor bank switch to position '1' and note down all the meter readings
5. Switch the capacitor bank switch to position '2' and note down all the meter readings
6. Bring back the capacitor bank switch to position 0 and switch off the motor by bringing auto transformer to 0V and remove the connections

**Tabular Column**

| Set value of C Input 1-phase AC(From MFM) | Sno | C( $\mu$ F) | Voltage (V) | Current (A) | Power (W) | Power F (COS $\theta$ ) No Units | Current in C ( $I_c$ ) (A) | XC= V/ $I_c$ ( $\Omega$ ) | Reactive power Q (VAR) | C  |
|-------------------------------------------|-----|-------------|-------------|-------------|-----------|----------------------------------|----------------------------|---------------------------|------------------------|----|
|                                           | 1   | 0           |             | $I_0 =$     |           |                                  | -----                      | -----                     | -----                  | -- |
|                                           | 2   | 8.5         |             | $I_1 =$     |           |                                  |                            |                           |                        |    |
|                                           | 3   | 17          |             | $I_2 =$     |           |                                  |                            |                           |                        |    |

**Calculations****THEORETICAL CALCULATIONS:**

Current in Capacitor  $C_1 = 8.5\mu\text{F}$  is  $I_{C1} = I_0 - I_1$

Current in Capacitor  $C_2 = 17\mu\text{F}$  is  $I_{C2} = I_0 - I_2$

Reactance of Capacitor  $X_c = V / I_c$

Reactive Power  $Q = I_c^2 * X_c$

The C value is calculated from above observation as  $C = \frac{1}{2\pi f X_c}$  where f is the frequency = 50Hz

Where C = capacitor in micro farads

**RESULT:** The power factor is observed to be improved by using capacitor.

**DISCUSSION OF RESULT:**

The power factor is observed to be improved by using capacitor

With  $C=0$  (without Capacitor), Power Factor is=\_\_\_\_\_

With  $C=8.5\mu\text{F}$ , Power Factor is=\_\_\_\_\_

With  $C=17\mu\text{F}$ , Power Factor is=\_\_\_\_\_

**EXPERIMENT-8****MAGNETISATION CURVE OF A SEPARATELY EXCITED D.C GENERATOR**

**AIM:** To obtain the magnetization characteristics (O. C. C) of a DC separately excited generator.

**EQUIPMENTS REQUIRED:**

| SL.NO | NAME OF EQUIPMENT | TYPE       | RANGE             | QUANTITY |
|-------|-------------------|------------|-------------------|----------|
| 1     | Ammeter           | M.C        | 0-1A              | 1        |
| 2     | Voltmeter         | M.C        | 0-300V            | 1        |
| 3     | Rheostat          | Wire wound | 1000 $\Omega$ /1A | 1        |

**NAME PLATE DETAILS:****DC Motor: (INPUT)****DC Separately Excited Generator:(OUTPUT)**

Voltage : -----

Voltage : -----

Current : -----

Current : -----

Power : -----

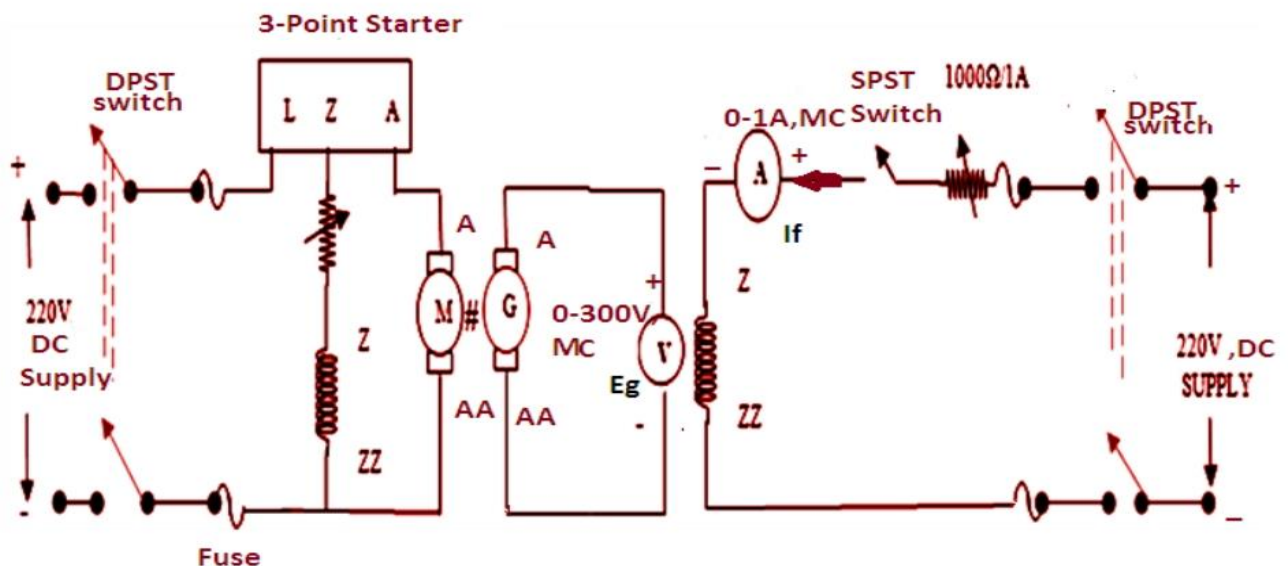
Power : -----

Speed : -----

Speed : -----

**THEORY**

An Electric generator consists of a rotor rotating in a magnetic field. The magnetic field may be produced by permanent magnets or by field coils. In the case of a machine with field coils, a current must flow in the coils to generate the field; otherwise no power is transferred to or from the rotor. The process of generating a magnetic field by means of an electric current is called *excitation*.

**CIRCUIT DIAGRAM:****PROCEDURE:**

1. Make the connections as per the diagram, choosing the meters suitable to the ratings of the machines.
2. Keep the motor field rheostat in the minimum position that of the generator field rheostat in the maximum position and start the motor by means of a starter.
3. Run the Motor at rated speed by varying field resistance.

## BASIC ELECTRICAL ENGINEERING LAB, EED

- Note down the voltage connected across A-AA of the Generator with switch SPST open. This e.m.f is due to the residual magnetism called Residual EMF.
- Keep the rheostat connected in field of generator side maximum, (So that field current  $I_f$  is minimum).
- Close the switch S1 and Increase the field current of the generator in steps by cutting out the resistance of the field rheostat.
- Note down the corresponding value of e.m.f induced by the generator.
- The resistance should be changed in one direction.
- The same procedure should be repeated with decreasing values of field current. Finally open SI switch and note down the value of e.m.f generated by residual magnetism.
- Draw the graph for the e.m.f generated Vs field current (both increasing and decreasing values)

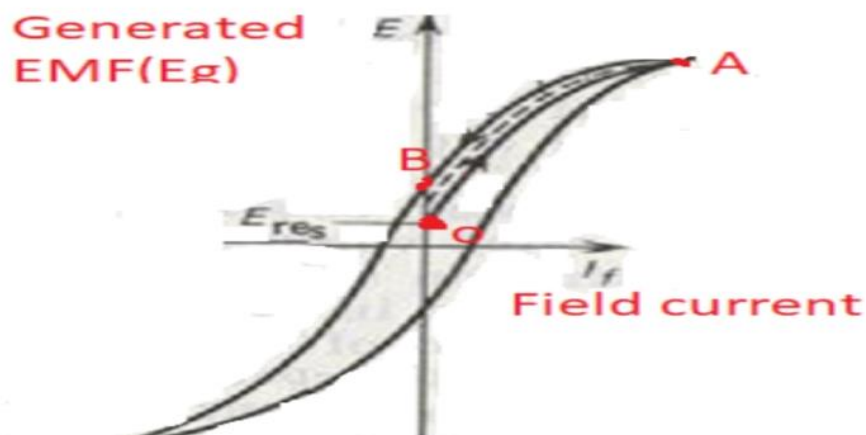
### **OBSERVATIONS:**

| S.NO | $I_f$<br>(A) | $E_o$ (INCREASING)<br>(V) | $E_o$ (DECREASING)<br>(V) | Average ( $E_o$ )<br>(V) |
|------|--------------|---------------------------|---------------------------|--------------------------|
|      |              |                           |                           |                          |
|      |              |                           |                           |                          |
|      |              |                           |                           |                          |
|      |              |                           |                           |                          |
|      |              |                           |                           |                          |

### **PRECAUTIONS:**

- All the connections should be clear and tight.
- The voltmeter and ammeter should be carefully chosen so that their ranges are more than max. Values to be measured.
- The field current should not be increased to cross rated value.

### **EXPECTED GRAPH:**

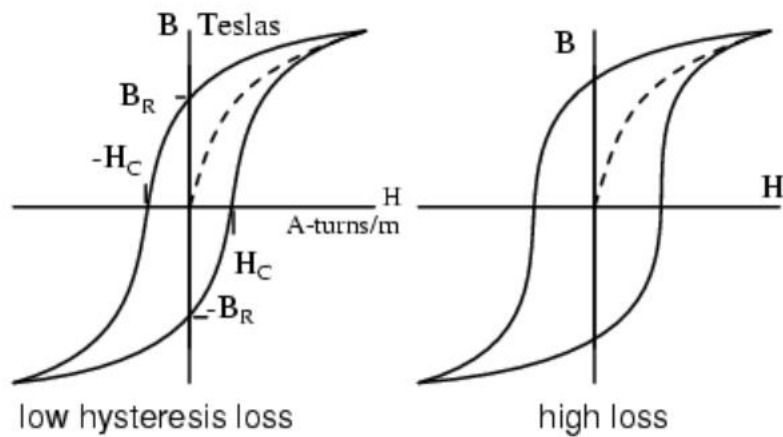


OA curve is  $E_g$  with increasing field current( $I_f$ )

BA curve is  $E_g$  with decreasing field current( $I_f$ ).

The area under these curves OA & BA is hysteresis loss(OBA)





**RESULTS:**

The Magnetization characteristics of separately excited DC generator at constant speed of  $N=1500\text{rpm}$  is plotted and hysteresis loss is observed.

**DISCUSSION OF RESULTS:**

The Residual EMF of separately excited DC generator with rating \_\_\_\_\_ is observed experimentally to be  $E_0 = \text{-----V}$

